

Caribbean Coral Reef Institute (CCRI) Water Quality Monitoring Lab University of Puerto Rico – Mayagüez Campus Department of Marine Sciences			
Document Title	Document Number	Version	
SOP Total Suspended Solids	WQP 008	1.0	

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1. Title

Standard Operating Procedure for Total Suspended Solids (TSS) Determination

2. Applicability

This SOP applies to determining TSS at shallow coral reef sites around Puerto Rico.

3. Purpose

- 3.1. Provide laboratory-specific instructions for the CCRI Water Quality Monitoring Lab personnel to measure coral reef seawater clarity by quantifying the TSS at samples from shallow coral reef sites around Puerto Rico.
- 3.2. Provide water quality data to be uploaded to the EPA WQX database for use by state and federal agencies and other parties interested in coral reef seawater TSS data following the Quality Assurance Project Plan.

4. Method

Standard Method 2540-D (Baird et al., 2017)

5. Health and Safety

- 5.1. Follow the safety indications for equipment and personnel suggested by the manufacturer.
- 5.2. Use heat-resistant gloves when working with the oven.

6. General Precautions

- 6.1. Always maintain two points of contact when deploying the Van Dorn bottle on the boat to avoid losing the bottle overboard.
- 6.2. Ensure the boat engines are idle and the captain approves of the Van Dorn bottle being deployed overboard before deploying the bottle to ensure that the rope does not become entangled with the boat propeller.
- 6.3. Do not hold samples for more than 48 h.

7. Training

- 7.1. Personnel shall be trained in the proper sample collection, handling, and analysis procedure for TSS.
- 7.2. Personnel shall receive training on how to complete the field sampling log for this SOP.

8. Required Materials

- 8.1. Van Dorn Bottle
- 8.2. Thermo Scientific PR305225G Drying Oven
- 8.3. Mettler Toledo ME204T Balance (220g ± 0.1mg)
- 8.4. Welch Vacuum Filtration Pump
- 8.5. Glass Vacuum Filtration Device

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- 8.6. Thermo Scientific Nalgene Acrylic Desiccator Cabinets (30.5cm)
- 8.7. Drierite Indicating Absorbent Pellets
- 8.8. Stir plate
- 8.9. Magnetic stir bar (10 mm)
- 8.10. Ready to Use pre-weighed Glass Microfiber Filter (RTU Filter)
- 8.11. 1 L Polypropylene Plastic Bottle for sampling (1)
- 8.12. Field Form (Appendix F)

9. Procedure

9.1. Sample Collection

- 9.1.1. Prepare a Van Dorn bottle to be deployed.
- 9.1.2. Lower the Van Dorn bottle in a slow and steady mode until it gets close to the bottom with caution not to touch the bottom and stir sediment up.
- 9.1.3. Drop the bottle messenger to close the Van Dorn at the desired depth.
- 9.1.4. Pull the Van Dorn bottle back up into the boat.
- 9.1.5. Connect the Tygon tubing to the Van Dorn bottle spigot to transfer water.
- 9.1.6. Push open the Van Dorn bottle's spigot valve to allow water flow.
- 9.1.7. Rinse the 1L plastic bottle for sample collection three times with water from the Van Dorn bottle.
- 9.1.8. Collect the sample until the bottle overflows.
- 9.1.9. Place the sample bottle inside the cooler until analysis in the lab.
- 9.1.10. Process samples within 48 hours.

9.2. Sample Processing

- 9.2.1. Bring samples to room temperature before analysis.
- 9.2.2. Record the weight (mg) of the preweighed RTU filter in the TSS analysis station lab notebook.
- 9.2.3. Assemble the glass vacuum filtration system.
- 9.2.4. Insert the preweighed RTU filter into the filter unit using tweezers.
 - 9.2.4.1. Ensure that the filter is oriented with the wrinkled side up.
- 9.2.5. Check the seawater sample for any large, heterogeneous floating particles such as *Sargassum* spp. fragments and exclude unless this is representative of the source waters.
 - 9.2.5.1. Record in the sample record whether any fragments were excluded.
- 9.2.6. Mix the seawater sample by inverting the bottle at least three times to homogenize the contents.
- 9.2.7. Transfer seawater from the bottle into a graduated cylinder, and record the volume in the TSS analysis station lab notebook.
 - 9.2.7.1. Selected sample volume should yield between 2.5 mg and 200 mg of dried residue.
 - 9.2.7.2. If filtration takes more than 10 min to complete, decrease sample volume.

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- 9.2.8. Pour the sample into the filtration system.
- 9.2.9. Turn on the pump to begin the suction on the glass filtration system.
- 9.2.10. After the sample volume is completely filtered, wash the filter with at least 10mL of reagent-grade water three times and allow all water to filter out.
 - 9.2.10.1. Allow complete drainage between rinses.
- 9.2.11. Carefully remove the filter from the filtration unit using tweezers from the edge without touching the sample and transfer it to the aluminum weighing dish.
- 9.2.12. Place the aluminum weighing dish with filter into the oven for at least 1 hour at 104°C to dry the filter.
- 9.2.13. After drying time is complete, remove the aluminum weighing dish from the oven using heat-resistant gloves.
- 9.2.14. Place the aluminum weighing dish inside the desiccator until it reaches ambient temperature ~15 minutes.
- 9.2.15. Weigh the filter and record the weight in the TSS analysis station lab notebook.
- 9.2.16. Repeat the drying, cooling, desiccating, and weighing cycle until a constant weight is obtained or until the weight change is less than 0.5 mg between replicate samples.
 - 9.2.16.1. Record each filter weight in the TSS analysis station lab notebook.

9.3. Calculation

$$\text{Total Suspended Solids (mg/L)} = \frac{(A - B) \times 1000}{\text{sample volume (ml)}}$$

where:

A = final weight of filter and dried residue (mg)

B = initial weight of pre-weighed filter (mg)

9.4. TSS Measurement Accuracy Verification

- 9.4.1. Prepare the TSS standard.
 - 9.4.1.1. Add 100 mg of Cellulose Type 20 or Celite 545 into a clean sample bottle.
 - 9.4.1.2. Measure 1000 mL of distilled water and add to the clean sample bottle with 100 mg cellulose.
 - 9.4.1.3. Shake the sample bottle.
 - 9.4.1.4. Place the standard solution on a stir plate with a stir bar for 15 minutes to thoroughly mix.
- 9.4.2. Prepare the TSS blank.
 - 9.4.2.1. Measure 1 L reagent water in a graduated cylinder and add to a clean sample bottle.
- 9.4.3. Perform the Sample Processing steps for the TSS sample and TSS blank as described in Section 9.2.
- 9.4.4. Perform the Calculation steps described in Section 9.3.

10. Documentation

- 10.1. Field data will be recorded in the Sampling Log (Appendix F).

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- 10.2. Filter weights will be recorded in the TSS analysis station lab notebook.
- 10.3. Field forms shall be kept in the designated file cabinet at the CCRI office.
- 10.4. Data contained in hard copy forms shall be digitized into the corresponding electronic spreadsheets as soon as possible after the task is completed.
- 10.5. All spreadsheets and depth profile logs should be uploaded to the internal CCRI Water Quality Database.
- 10.6. All data will be electronically archived and backed up online.

11. Cleaning and Maintenance

Clean equipment thoroughly with water after each use.

12. Troubleshooting

N/A

13. QA/QC

- 13.1. As per QAPP Table 4, seawater TSS should be measured in milligrams per liter (mg/L) with an accuracy of ± 5 mg/L ($\pm 20\%$) and a precision of $<20\%$.
- 13.2. The accuracy of total suspended solids will be verified by completing the Sample Processing and Calculation steps with a 100 mg/L internal standard and a blank every 20 samples.
 - 13.2.1. The standard should be within $\pm 20\%$ of measurement and the blank is reported as the minimum detection level for TSS.
- 13.3. In accordance with the QAPP, 10% of all samples will be measured in duplicate to ensure the precision of replicate measurements is $<20\%$.

14. References

- 14.1. Baird, R. B., Eaton, A. D., & Rice, E. W. (2017). Standard methods for the examination of water and wastewater (Vol. 23). E. W. Rice (Ed.). Washington, DC: American Public Health Association.